Minh Nam TRAN

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Motivated and technically skilled graduate student seeking an internship in machine learning and deep learning, with hands-on experience in AI, Deep Learning, and Natural Language Processing.

Education

Purdue University *Aug. 2025 – Present*
Ph.D. Student in Computer Science *West Lafayette, USA*

- Current GPA: 4.00/4.00
- Courses: Algorithm Design, Analysis, and Implementation, Data Communication and Computer Networks, and Foundations Of Deep Learning.

University of Science - VNUHCM *Sep. 2020 – Oct. 2024*
B.Sc. in Computer Science, Advanced Program *Ho Chi Minh, Vietnam*

- GPA: 3.98/4.00 - Degree class: Excellent.
- Thesis title: "*Exploring and Improving Language Understanding Abilities of Vietnamese Language Models*," advised by Prof. Dien Dinh and Dr. Long HB Nguyen (score: 10.0 /10.0)

Experience

Saparov's Lab - Purdue University *Aug. 2025 - Present*
Graduate Research Assistant *West Lafayette, US*

- Conducting research in multi-agent systems, language model's logical thinking, and applications of language models under supervision of Prof. Abulhair Saparov.
- Engaging in two research projects, including "Multi-agent code reasoning with Vision Language Models for graphic design editing" and "Verifying Language Model's outputs in scientific and technical analysis using multi-agent system."

Computational Linguistics Center - VNUHCM *May 2022 – Aug. 2025*
Undergraduate Research Assistant *Ho Chi Minh, Vietnam*

- Conducted research under supervision of Prof. Dien Dinh and Dr. Long Hong Buu Nguyen.
- Investigated in evaluating Vietnamese language models and finally published a Vietnamese general language understanding evaluation benchmark (ViGLUE) with findings about the pre-trained language models, using both few-shot learning and direct fine-tuning.
- Designed the dual relationship learning method for improving language models on natural language inference tasks in Vietnamese medical domain and outperformed baseline on ViMedNLI and $\text{ViNLI}_{\text{health}}$ datasets.

Viettel Group *Apr. 2023 – Oct. 2023*
AI Intern, Viettel Digital Talent 2023 *Hanoi, Vietnam*

- Completed two projects in machine learning/deep learning under the supervision of Dr. Nguyen Van Nam: the table-to-text task and the text-to-SQL problem with large language models.
- Fine-tuned encoder-decoder Transformer models, including BART and T5, for table-to-text generation with synthesized data created by gpt3.5.
- Applied the QLoRA method to fine-tune CodeLLama-7B and CodeLlama-14B models on a single H100 GPU for the text-to-SQL problem, achieving the performance at 72.7% execution accuracy and 61.5% exact match on the test subset of Spider benchmark.

- Deployed text2sql models under 4-bit quantization on the Viettel Machine Learning Platform system with FastAPI, Docker, and `Llama-cpp`.

Users and Information Lab, KAIST *Jun. 2023 – Aug. 2023*
Visiting Research Student *Daejeon, South Korea*

- Worked under the supervision of Prof. Alice Oh and PhD. student Rifki Afina Putri.
- Investigated adapting the BLOOM model family to four Vietnamese downstream tasks using low-rank adaptation with instruction tuning and improved performance using the length-penalized loss.

Selected Publications

1 Tuan-An To, **Minh-Nam Tran**, Trong-Bao Ho, Thien-Loc Ha, Quang-Tan Nguyen, Hoang-Chau Luong, Thanh-Duy Cao, and Minh-Triet Tran. 2024. Multi-perspective traffic video description model with fine-grained refinement approach. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 7075–7084. DOI: 10.1109/CVPRW63382.2024.00701.

2 **Minh-Nam Tran**, Phu-Vinh Nguyen, Long Nguyen, and Dinh Dien. 2024. ViGLUE: a Vietnamese general language understanding benchmark and analysis of Vietnamese language models. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 4174–4189. DOI: 10.18653/v1/2024.findings-naacl.261.

3 **Minh-Nam Tran**, Phu-Vinh Nguyen, Long Nguyen, and Dien Dinh. 2024. ViMedAQA: a Vietnamese medical abstractive question-answering dataset and findings of large language model. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)*. Xiyan Fu and Eve Fleisig, editors. Association for Computational Linguistics, Bangkok, Thailand, (Aug. 2024), 252–260. ISBN: 979-8-89176-097-4. DOI: 10.18653/v1/2024.acl-srw.31.

Skills & Technologies

- **Programming:** Python, C/C++, Java, R, Shell, JavaScript.
- **Frameworks:** NumPy, PyTorch, TensorFlow, Scikit-learn, FastAPI, llama-cpp, LangChain, LangGraph, Model Context Protocol.
- **Tools and Technology:** Git and GitHub, Docker, Linux server, Slurm, LaTex.
- **Specialized skills:** Data Science, Deep Learning, Natural Language Processing, Large Language Modeling, Vision Language Modeling, Generative AI, LLM Distributed Training.

Training & Relevant Coursework

TensorFlow Developer Certificate *Jul. 2023*
by TensorFlow

Deep Learning *Jun. 2022*
by DeepLearning.AI

References

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